

Self-Evolving Agentic AI & Trading - Intact System Blueprint 2026

Expanded research on self-evolving, reflective, meta-learning agentic AI and trading systems.

Core: Agents that capture experience (successes/failures/regret), reflect in structured form (lessons, violated assumptions), and update memory, tools, workflows, rules, and even the improvement process itself.

Key 2026 work:

- TradingGroup (arXiv 2508.17565): Multi-agent with per-agent self-reflection (forecasting, style, decision) that distills past successes/failures for analogous scenarios + dynamic risk (adaptive SL/TP/sizing).
- MetaAgent (2508.00271): Starts minimal, evolves via self-reflection + verified reflection + dynamic context engineering + autonomous in-house tool construction ("meta tool learning") without retraining.
- TradingAgents: Realized returns + alpha trigger LLM reflections that are injected into future prompts/memory.
- Broader surveys: What to evolve (memory/tools/architecture/workflows), when (intra vs inter-task), how (textual feedback, evolutionary search, meta-learning).

Diagnosis of current system (tredo/ATES):

Strong foundation exists (hierarchical groups, emerging debate with trained memory recall, rich episodes + PostTradeReflection containing regret/lessons/suggested changes, DisciplinedCore in Rust, temporal loops, Reflector with deep reflection, skills).

Critical gaps making it "not working" for full self-evolving autonomy: Execution and backtester are stubs/placeholders; debate not fully end-to-end with aggregator; reflection not reliably closing the loop into meta-control and procedural memory/rule adaptation; no automatic outcome resolution + reflection injection on future similar runs; missing realistic LOB simulation for trustworthy adaptation signals; multi-asset and dynamic risk adaptation incomplete.

The Intact System Blueprint (complete target):

Build directly on existing strengths. Add the missing closed self-evolution loop:

Perception (multi-asset verified feeds) -> Specialists + deterministic Skills -> Full Debate (Proposer/Critic/Risk/Historian with recall) + Aggregator -> Decision + Dynamic Risk (memory-informed adaptive parameters) -> Realistic Execution (paper with LOB impact/slippage first, then live) -> Rich Episode capture -> Structured Reflection (with prior memory context first) -> Memory update (episodic + vector + procedural lessons) -> Meta layer (high-regret patterns trigger concrete rule/risk/tool/prompt adaptations, with safety gates) -> Guardian (hard deterministic kill-switches and checks that never get evolved away).

Self-evolution loop made intact and measurable: Action + full trace -> realistic outcome -> reflection (regret + lessons) -> memory/procedural update -> meta proposes updates -> gated apply -> future similar situations retrieve improved state -> lower regret / better risk-adjusted results on analogous setups.

Enablers: Realistic multi-asset simulator (QuantReplay-style or equiv), persistent decision log + automatic realized return resolution + reflection injection, versioned auditable adaptations, full COT + adaptation history, asset-class aware + cross-asset risk.

Safety: All hard guardrails and kill switches remain in deterministic code outside the evolving intelligence layers.

Prioritized path to intact working system: Complete debate end-to-end, make execution real + realistic paper modeling, wire closed outcome->reflection->meta-update loop, integrate proper simulator, add self-evolution metrics, expand multi-asset.

This gives a complete, highly adaptive, self-improving autonomous agentic trading system for stocks, crypto and multi-markets that systematically learns from mistakes and updates itself.

References: TradingGroup, MetaAgent, Self-Evolving Agents Survey (2507.21046), TradingAgents reflection mechanisms, Reflexion, HyperAgents, internal tredo architecture (episodes/regret/meta/debate/DisciplinedCore).

Not financial advice.